Wen Li

Ph.D. Candidate, University of Chicago
MRS GSA Gold Award | Benjamin Ball Freud Merit Scholar
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Wen Li is a sixth-year Ph.D. candidate at the University of Chicago, advised by Professors Bozhi Tian and Pedro Lopes. His work bridges biomedical engineering and human–computer interaction (HCI). He developed wearable electrochemical biointerfaces for skin protection, wearable electrical devices for electrotactile-based thermal referral and wearable haptic input/output devices that model and personalize human tactile and thermal perception. He connects fundamental science with computer science, linking biointerface engineering with computational modeling and human-centered design. He has published in leading journals and HCI venues, including *Nature*, *Nat. Chem. Eng.*, *Nat. Nanotechnol.*, *Nat. Biomed. Eng.*, *Nat. Sustain.*, *Matter*, *Device*, and *ACM UIST*. His ongoing work is under review at *Sci. Robot.*. His work was recognized with a double winning of 2025 **MRS Graduate Student Gold Award** and **Best Poster Award**.

Education

Ph.D., The University of Chicago, Chicago, IL, USA **09/2021 – Present (exp. 03/2027)**
Advisors: **Bozhi Tian (Chemistry)** and **Pedro Lopes (Computer Science)**.
Selected courses: Solids, Materials, and Surfaces; Engineering Interactive Electronics onto PCB; Electronic Instrumentation in Biophysics.

Exchange Student, UC Berkeley, Berkeley, CA, USA **01/2020 – 06/2020**
Advisor: Prof. Phillip B. Messersmith. Major GPA: 4.0/4.0

B.S. in Chemistry with Honors, Nanjing University, Nanjing, Jiangsu, China **09/2017 – 06/2021**
Thesis advisor: Prof. Cheng-Hui Li. Major GPA: 4.659/5.000; Rank: 1/79.

Research Experience

Engineering Bioelectrical Interfaces for Therapy and Perception, UChicago **Chicago, IL**
advisor: Prof. Bozhi Tian (Chemistry), Prof. Pedro Lopes (Computer Science) **09/2021 – Present**

(a) Contributed to the fabrication of **porous silicon** photoelectrochemical devices and **photo-stimulation** experiment in vitro on cardiomyocytes, ex vivo on heart tissue, and in vivo on rat hearts. (*Nature*, 2024; *Nat. Nanotechnol.*, 2025; *Nat. Biomed. Eng.*, 2026)
Skills: **Optical stimulation**, **Electrophysiology**, **Semiconductor characterization**

(b) Developed a wearable wireless electrochemical device to sustainably and programmably deliver H_2 for ischemia-reperfusion heart disease and pressure ulcer (bedsore) treatment. (*Nat. Chem. Eng.*, *2025*)
Skills: **Printed Circuit Board (PCB) design**, **Electrochemistry**, **Embedded System**

(c) Developed a **wearable haptic input/output device** that uses **Bayesian machine learning** to learn and render personalized thermal and tactile percepts for **VR**, haptic sensory substitution, and **haptic-guided robot teleoperation**, by combining **electrotactile** stimulation with **thermal referral**. (*ACM UIST*, *2026*; *Under review at Science Robotics, 2026*)
Skills: **Tactile sensing**, **Bayesian ML**, **Robot Teleoperation, PCB design, Embedded System**

Self-healing polymers by dynamic coordination chemistry, Nanjing University **Nanjing, China**
advisor: Prof. Cheng-Hui Li **09/2018 – 06/2021**
Developed a new polymeric material by crosslinking small molecule units with metal ions. The material simultaneously possesses high mechanical properties, recyclability, and self-healing. (*CCS Chem.*, 2022)
Skills: **Polymer synthesis**, **Polymer characterization**

Working Experience

Materials Research Intern, Integrated Circuit Lab, Futurewei, San Jose, CA **06/2024 – 09/2024**
(a) Developed a group of polymers for **Meta-optics** with tunable refractive indices in nanometer scale.
(b) Developed self-healing polymer coatings for electronics and microchip **radiative cooling**.
(c) Delivered **two lead-author patents** and **two co-authored papers**. Won **Outstanding Intern Award** in Futurewei Summer 2024 intern research project (3 awardees out of 30-40 interns).

Teaching Assistant, UChicago, Chicago, IL **09/2021 – 06/2022, 09/2026-present**
Served as TA for four classes: Organic Chemistry, Physical Chemistry experiment, Chemical Kinetics, and Honor General Chemistry. Developed and delivered discussion classes once a week, supervised over 30 students in teaching laboratories and writing laboratory reports, and held weekly office hours.

Academic & Community Service

Cofounder of HPad, College New Venture Challenge (CNVC), The University of Chicago
Built a startup developing a medical device for bedsore treatment based on my Nature Chemical Engineering work. Selected as one of 15 teams out of ~70 to advance to CNVC Phase II; secured $5K in seed funding.

Team Leader in Academic Service Department, Nanjing University
Held academic discussion seminars and final exam review seminars. Won the Exceptional Team Leader in Academic Service Department and the Exceptional Freshman Mentor and Friend awards.

Reviewer for journals and conferences
Reviewed 10+ research manuscripts (ACS Nano, Device, IMWUT, EuroHaptics, IEEE Trans. Vis. Comput. Graph.).

Awards

Award for Outstanding Students Abroad, ***China's Ministry of Education***	**2026**
Graduate Student Gold Award, ***Materials Research Society***	**2025**
Best Poster Award, ***Materials Research Society***	**2025**
The Olshansky Graduate Student Travel Award, ***The University of Chicago***	**2025**
Benjamin Ball Freud Merit Scholarship, ***The University of Chicago***	**2025**
Outstanding Intern Award, ***Futurewei Technologies, Inc.***	**2024**
Entrepreneurship for Science and Medicine, ***UChicago Booth***	**2023**
McCormick Fellowship, ***The University of Chicago***	**2021**
Merit Student of Jiangsu Province, **Jiangsu Province**	**2020**
Excellent Student of Nanjing University, **Nanjing University**	**2019**
National Scholarship, **China's Ministry of Education**	**2018**

Selected Publications and Patents (*=corresponding author)

Full publication available at Google Scholar: https://scholar.google.com/citations?user=9D4DmSUAAAAJ&hl=en

1. **W Li***, E Adam, Z Xu, J Yao, A Esmer, X Ni*, B Tian*, P Lopes*. A wearable input/output interface learns personalized models of haptic perception. Under review in ***Science Robotics***, **2026**. Preprint available here.
2. **W Li***, R Ni, B Tian, P Lopes. Electrotactile Improves Thermal Referral. ***UIST*, 2026**. Video here.
3. **W Li***, J Zhang, et al., B Tian*, Hydrogen Evolution in Portable Hydrogel Electrochemical Bioelectronics for Ischemia-Reperfusion Therapy. ***Nature Chemical Engineering***, 2, 484-497, **2025**.
   News & Views: Hydrogen therapy for ischemic injuries. ***Nature Chemical Engineering***, 2, 467-469, **2025**.
4. **W Li**, J Huberman-Shlaes, B Tian*. Perspectives on Multiscale Colloid-Based Materials for Biomedical Applications. ***Langmuir***, 39, 13759-13769, **2023**.
5. **W Li**, HQ Wang, et al., CH Li*. A strong and rigid coordination adaptable network that can be reprocessed and recycled at mild conditions. ***CCS Chemistry***, 4, 3781-3797, **2022**.
6. P Li, J Zhang, H Hayashi, J Yue, **W Li**, et al., B Tian*. Monolithic silicon for high spatiotemporal translational photostimulation. ***Nature,*** 626, 990-998, **2024**.
7. J Zhang*, P Li, J Yue, LY Meng, **W Li**, et al., B Tian*. Gold-modified nanoporous silicon for photoelectrochemical regulation of intracellular condensates. ***Nature Nanotechnology***, 20, 835-844, **2025**.
8. C Müller, K Klompmaker, Y Zhang, et al., **W Li**, et al., B Tian, MD Dong, ML Chen*. Biomimetic graphitic carbon nitride nanoparticles for multiscale photomodulation and therapeutic intervention. ***Nature Biomedical Engineering***, **2026**.
9. C Yang, P Li, et al., **W Li**, et al., B Tian*. A bioinspired permeable junction approach for sustainable device microfabrication. ***Nature Sustainability***, 7, 1190-1203, **2024**.
10. C Sun, J Yue, P Li, **W Li**, J Ma, C Yang, S Kim, B Tian*. Sustainable conversion of husk into viscoelastic hydrogels for value-added biomedical applications. ***Matter***, 8, 3, **2025.**
11. LY Meng, J Zhang, P Li, **W Li**, et al., B Tian*. Scalable Bioelectronic Production of Extracellular Vesicles for Cardiac Tissue Regeneration. **Device**, 3, 100968, **2025**.
12. **W Li**, Y Cao, J Wang, J Han, P Wu. Polymeric material and method of forming the same. **US Patent App.** No. PCT/US2024/058461, **2024**.
13. **W Li**, Y Cao, Z Li, J Han, J Wang, P Wu, L Peng. Polymer layer stack and methods of forming thereof. **US Patent App.** No. PCT/US2024/058694, **2024**.